

EXPERIMENT 11

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Question 1:

Question:

A company stores the monthly salary history of its employees in an EmployeeSalaryHistory table.

Write an SQL query to find the employees whose salary has increased in every recorded month compared to the immediately previous month. For each salary record, compare the current month's salary with the employee's previous month's salary using a Window Function. Display only those employees for whom all later salary records are greater than their respective previous salary.

Output should display:

- Employee ID
- Employee Name

Answer:

```
SELECT DISTINCT employee_id, employee_name
```

```
FROM (
```

```
  SELECT
```

```
    employee_id,
```

```
    employee_name,
```

```
    salary,
```

```
    LAG(salary) OVER (
```

```
      PARTITION BY employee_id
```

```
      ORDER BY salary_month
```

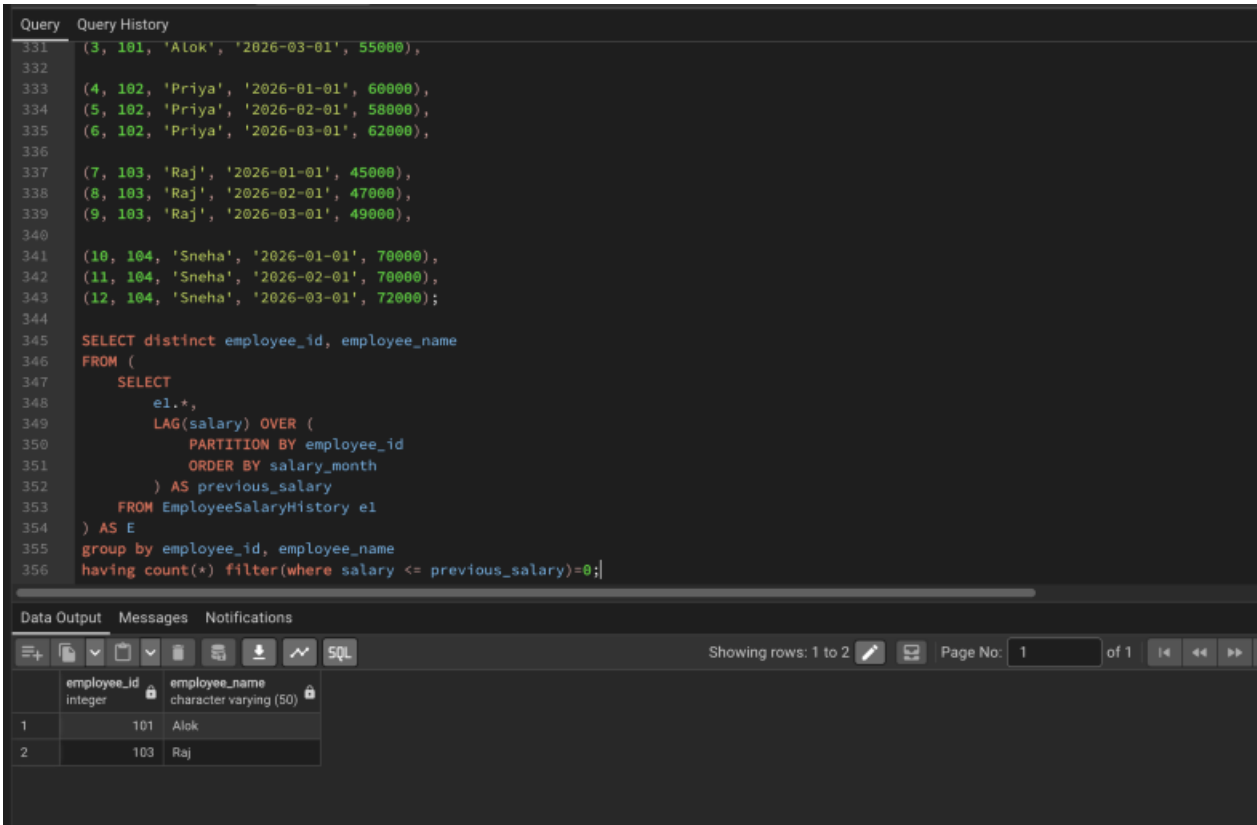
```
    ) AS previous_salary
```

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```

FROM EmployeeSalaryHistory
) AS E
GROUP BY employee_id, employee_name
HAVING COUNT(*) FILTER (
    WHERE salary <= previous_salary
) = 0;

```



The screenshot shows a SQL IDE with a query window and a data output window. The query window contains the following SQL code:

```

331 (3, 101, 'Alok', '2026-03-01', 55000),
332
333 (4, 102, 'Priya', '2026-01-01', 60000),
334 (5, 102, 'Priya', '2026-02-01', 58000),
335 (6, 102, 'Priya', '2026-03-01', 62000),
336
337 (7, 103, 'Raj', '2026-01-01', 45000),
338 (8, 103, 'Raj', '2026-02-01', 47000),
339 (9, 103, 'Raj', '2026-03-01', 49000),
340
341 (10, 104, 'Sneha', '2026-01-01', 70000),
342 (11, 104, 'Sneha', '2026-02-01', 70000),
343 (12, 104, 'Sneha', '2026-03-01', 72000);
344
345 SELECT distinct employee_id, employee_name
346 FROM (
347     SELECT
348         e1.*,
349         LAG(salary) OVER (
350             PARTITION BY employee_id
351             ORDER BY salary_month
352         ) AS previous_salary
353     FROM EmployeeSalaryHistory e1
354 ) AS E
355 group by employee_id, employee_name
356 having count(*) filter(where salary <= previous_salary)=0;

```

The data output window shows the following results:

employee_id	employee_name
1	101 Alok
2	103 Raj

Question 2:

Question:

A company maintains employee details in an Employee table and department information in a Department table.

Write an SQL query to find the employees whose salaries belong to the **top three distinct salary levels** within their respective departments.

If multiple employees have the same salary, they should receive the same salary rank and all of them should be included in the result.

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Output should display:

- Department Name
- Employee Name
- Salary

Answer:

SELECT

D.name AS Department,

E.name AS Employee,

E.salary

FROM (

SELECT

*,

DENSE_RANK() OVER (

PARTITION BY departmentId

ORDER BY salary DESC

) AS rnk

FROM Employee

) AS E

JOIN Department D

ON E.departmentId = D.id

WHERE rnk <= 3;

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```

397
398 --Solution:
399 SELECT D.name AS Department,
400        E.name AS Employee,
401        E.salary
402 FROM (
403     SELECT *,
404            DENSE_RANK() OVER (
405                PARTITION BY departmentId
406                ORDER BY salary DESC
407            ) AS rnk
408     FROM Employee
409 ) E
410 JOIN Department D
411 ON E.departmentId = D.id
412 WHERE rnk <= 3;
413
414

```

Data Output Messages Notifications

	department character varying (50)	employee character varying (50)	salary integer
1	IT	Max	90000
2	IT	Jim	85000
3	IT	John	85000
4	IT	Alex	75000
5	Sales	Henry	80000
6	Sales	Susan	75000
7	Sales	David	70000
8	HR	Tina	60000